



GOSFORD HIGH SCHOOL

2010 HIGHER SCHOOL CERTIFICATE

MATHEMATICS EXTENSION 2

ASSESSMENT TASK 2

PART A

Time Allowed – 60 minutes

+ 5 minutes reading time

All necessary working should be shown.

Full marks may not be awarded for unnecessarily untidy work or work that is poorly organized.

Students must begin each new question on a new page.

Questions will be collected separately at the conclusion of the assessment task.

All questions are to be attempted.

GRAPHS (21 marks)

a) (i) Sketch $y = x^3 + 1$ (1)

(ii) Hence, on the **same** graph, sketch $y = \sqrt{x^3 + 1}$ (2)

b) (i) Sketch $y = 2^x - 1$ (1)

(ii) Hence, on the **same** graph, sketch $y = \frac{1}{2^x - 1}$ (2)

c) (i) Sketch $y = 1 - |x|$ (1)

(ii) Hence, on a **new** graph, sketch $|y| = 1 - |x|$ (2)

d) $f(x) = 2 \cos x$ where $-\pi \leq x \leq \pi$.
Sketch $y = f(x)$ and $y = [f(x)]^3$ on the **same** number plane. (4)

e) Sketch $y = x \sin x$ for $-2\pi \leq x \leq 2\pi$ (4)

f) Sketch the curve $y = \frac{x^2 + 3x}{x + 1}$, showing all intercepts with the co-ordinate axes and asymptotes. (4)

COMPLEX NUMBERS (21 marks)

- a) Let $z = 3 + 2i$ and $w = 1 - 5i$.

Find, in the form $x + iy$,

(i) zw (ii) $\frac{w}{\bar{z}}$ (2)

- b) (i) Express $-1 + i$ and $1 + i\sqrt{3}$ in modulus-argument form. (2)

(ii) Hence express $\frac{(-1 + i)^4}{(1 + i\sqrt{3})^5}$ in the form $x + iy$. (3)

- c) Sketch the following locus on separate Argand diagrams

(i) $-1 \leq \text{Im}(z) < 2$ (2)

(ii) $\arg(z - i) = \arg(z + 1)$ (2)

d) (i) Sketch $|z - 1 - i\sqrt{3}| = 2$, (2)

(ii) Hence find the range of values of $\text{Arg}(z)$ (2)

e) Describe in simple geometric terms the locus of z if $\frac{1}{2}(z + \bar{z}) = |z| - 2$ (3)

f) Given $z = \frac{1 - \cos 2\theta + i \sin 2\theta}{1 + \cos 2\theta - i \sin 2\theta}$, show that z is purely imaginary (3)

TABLE OF STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note $\ln x = \log_e x, \quad x > 0$



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MATHEMATICS EXTENSION 2

ASSESSMENT TASK 2

PART B

Time Allowed – 60 minutes

+ 5 minutes reading time

All necessary working should be shown.

Full marks may not be awarded for unnecessarily untidy work or work that is poorly organized.

Students must begin each new question on a new page.

Questions will be collected separately at the conclusion of the assessment task.

All questions are to be attempted.

INTEGRATION (20 marks)

a) Find

(i) $\int \frac{e^{\tan x}}{\cos^2 x} dx$ (1)

(ii) $\int \tan^4 x \, dx$ (3)

(iii) $\int x \cos(2x) dx$ (2)

(iv) $\int \frac{dx}{\sqrt{x^2 - 8x + 25}}$ (2)

b) Evaluate

(i) $\int_0^1 \sqrt{4 - x^2} dx$ using the substitution $x = 2 \sin \theta$ (4)

(ii) $\int_2^{10} \frac{x^2 dx}{\sqrt{x-1}}$ (4)

c) (i) Show that $\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$, (2)

(ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \sin(5x) \cos(3x) dx$ (2)

POLYNOMIALS (22 marks)

- a) Factorise $x^4 - x^2 - 20$ over the
- (i) Real field (R) (1)
- (ii) Complex field (C). (1)
- b) If $2 + i$ is a root of the polynomial $S(x) = x^3 - 2x^2 - 3x + 10$, determine the other two roots. (2)
- c) Find the roots of the polynomial $x^4 - 6x^3 + 12x^2 - 10x + 3$ given that it has a root of multiplicity 3 (3)
- d) The equation $x^3 - 5x^2 + 8x + 24 = 0$ has roots α , β and γ .
Find the polynomial equation with roots α^2 , β^2 and γ^2 (3)
- e) Resolve $\frac{2x^2 + x + 5}{(x-3)(x^2+4)}$ into partial fractions over the Real field. (4)
- f) Find all the zeros of the polynomial $\phi(x) = 2x^3 - 5x^2 + x + 3$, given that it has one rational zero. (3)
- g) (i) If $z = \cos \theta + i \sin \theta$, show that
 $z + z^{-1} = 2 \cos \theta$ and that $z^n + z^{-n} = 2 \cos(n\theta)$ (2)
- (ii) By expanding $\left(z + \frac{1}{z}\right)^4$, find an expression for $\cos^4 \theta$ in the form
 $A \cos(4\theta) + B \cos(2\theta) + C$ (3)

TABLE OF STANDARD INTEGRALS

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$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

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$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

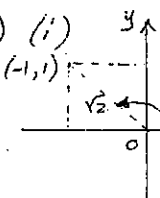
Note $\ln x = \log_e x, \quad x > 0$

COMPLEX NUMBERS

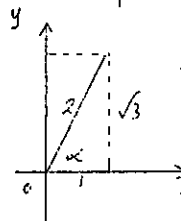
a) (i) $zw = (3+2i)(1-5i)$
 $= 3 - 15i + 2i + 10$
 $= 13 - 13i$

(ii) $\frac{w}{z} = \frac{1-5i}{3-2i} \times \frac{3+2i}{3+2i}$
 $= \frac{13-13i}{9+4}$
 $= \frac{13-13i}{13}$
 $= 1-i$

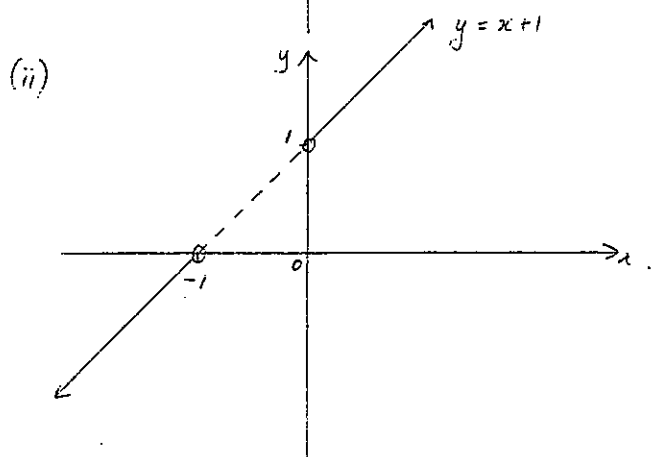
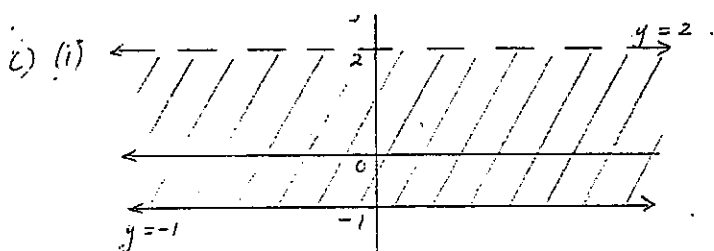
b) (i) $-1+i = \sqrt{2} \operatorname{cis} \frac{3\pi}{4}$
 $= \sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$



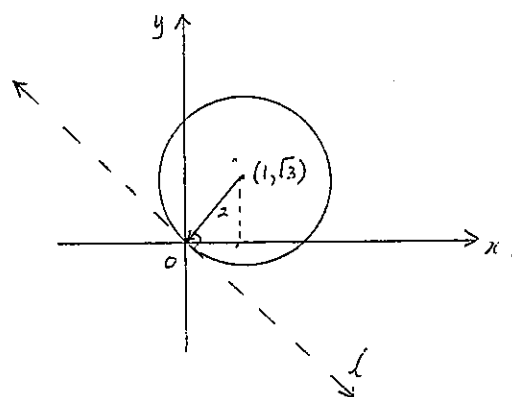
$1+i\sqrt{3} = 2 \operatorname{cis} \frac{\pi}{3}$
 $= 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$



(ii) $\frac{(-1+i)^4}{(1+i\sqrt{3})^5} = \frac{(\sqrt{2} \operatorname{cis} \frac{3\pi}{4})^4}{(2 \operatorname{cis} \frac{\pi}{3})^5}$
 $= \frac{4 \operatorname{cis} 3\pi}{32 \operatorname{cis} \frac{5\pi}{3}}$
 $= \frac{1}{8} \operatorname{cis} \left(\frac{4\pi}{3} \right)$
 $= \frac{1}{8} \left[\cos \left(-\frac{2\pi}{3} \right) + i \sin \left(-\frac{2\pi}{3} \right) \right]$
 $= \frac{1}{8} \left[-\frac{1}{2} - i \times \frac{\sqrt{3}}{2} \right]$
 $= -\frac{1}{16} (1+i\sqrt{3})$
 $= -\frac{1}{16} - \frac{i\sqrt{3}}{16}$



d) (i)



(ii) Gradient of line (L) = $-\frac{1}{\sqrt{3}}$
 $\therefore -\frac{\pi}{6} < \operatorname{Arg}(z) < \frac{5\pi}{6}$

e) $z = x+iy$

$\therefore \frac{1}{2} (2x) = \sqrt{x^2+y^2} - 2$

$x+2 = \sqrt{x^2+y^2}$

$x^2+4x+4 = x^2+y^2$

$y^2 = 4x+4$

Parabola with vertex $(-1, 0)$ and
 A.C. $u = 0$

$$f) \sin 2\theta = 2\sin\theta\cos\theta$$

$$\cos 2\theta = 2\cos^2\theta - 1 \text{ and } \cos 2\theta = 1 - 2\sin^2\theta.$$

$$1 + \cos 2\theta = 2\cos^2\theta \quad 2\sin^2\theta = 1 - \cos 2\theta.$$

$$z = \frac{2\sin^2\theta + 2i\sin\theta\cos\theta}{2\cos^2\theta - 2i\sin\theta\cos\theta}.$$

$$= \frac{2\sin\theta(\sin\theta + i\cos\theta)}{2\cos\theta(\cos\theta - i\sin\theta)}$$

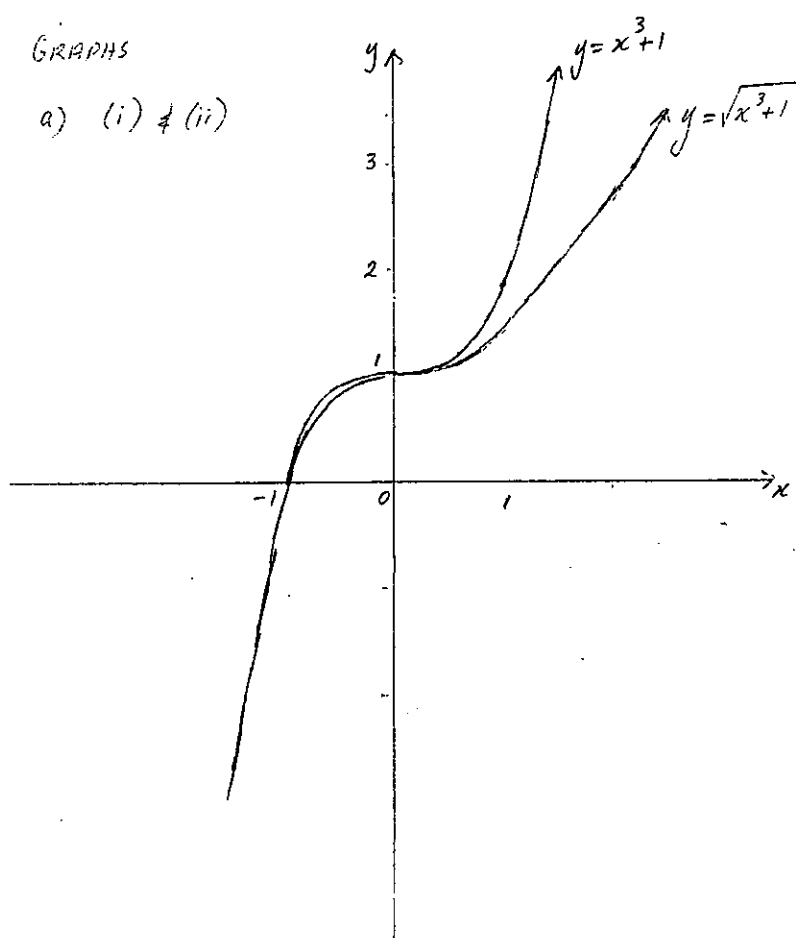
$$= \frac{2i\sin\theta(\cos\theta - i\sin\theta)}{2\cos\theta(\cos\theta - i\sin\theta)}$$

$$= i\tan\theta.$$

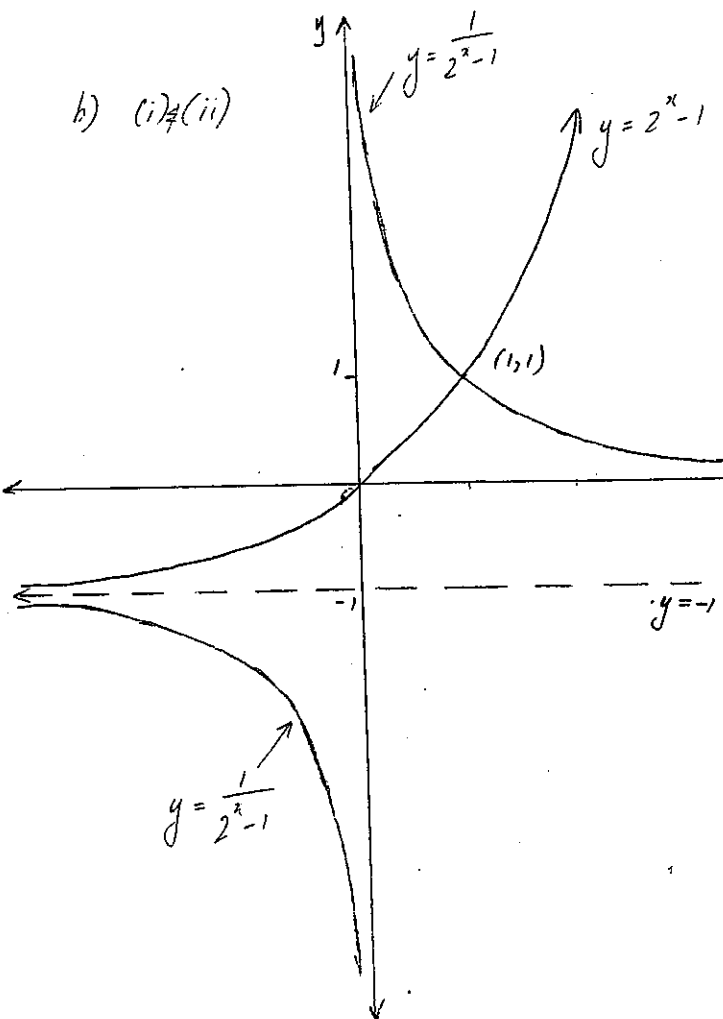
$\therefore z$ is purely imaginary.

GRAPHS

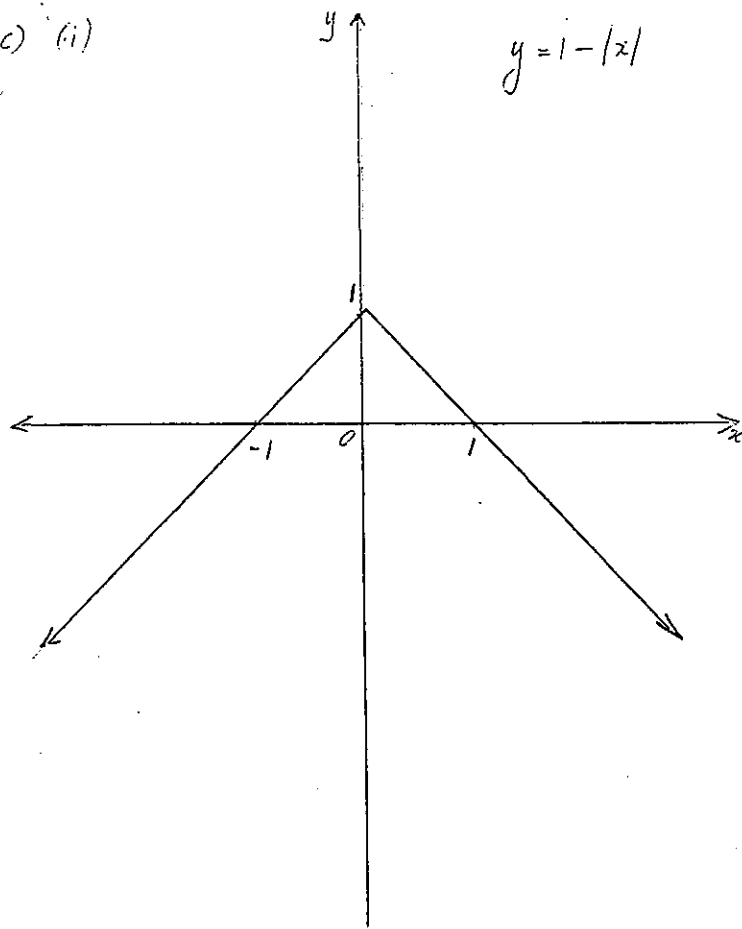
a) (i) & (ii)



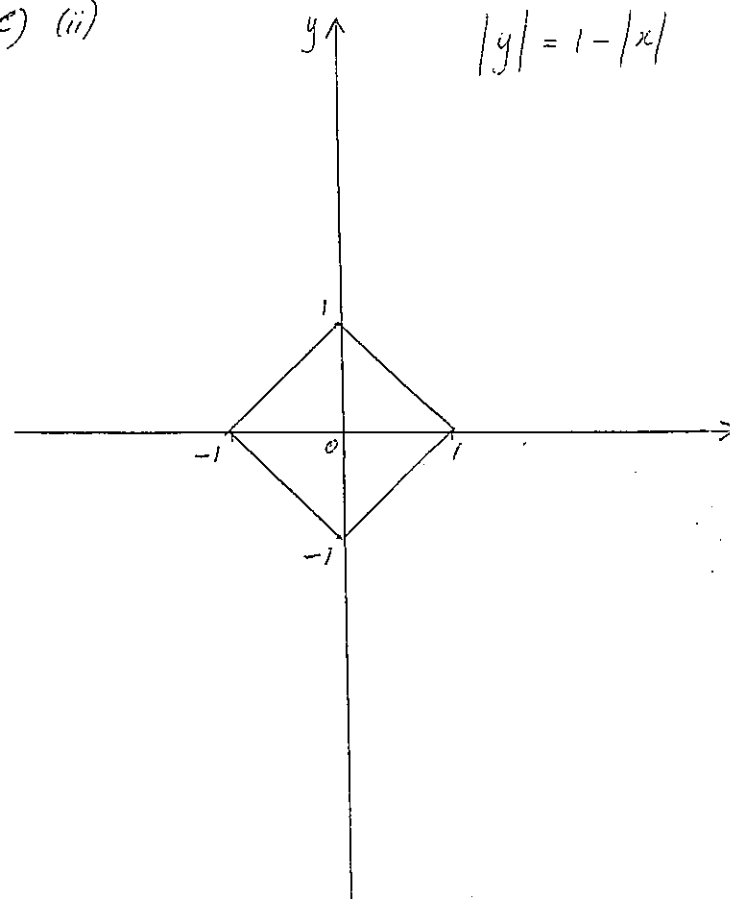
b) (i) & (ii)



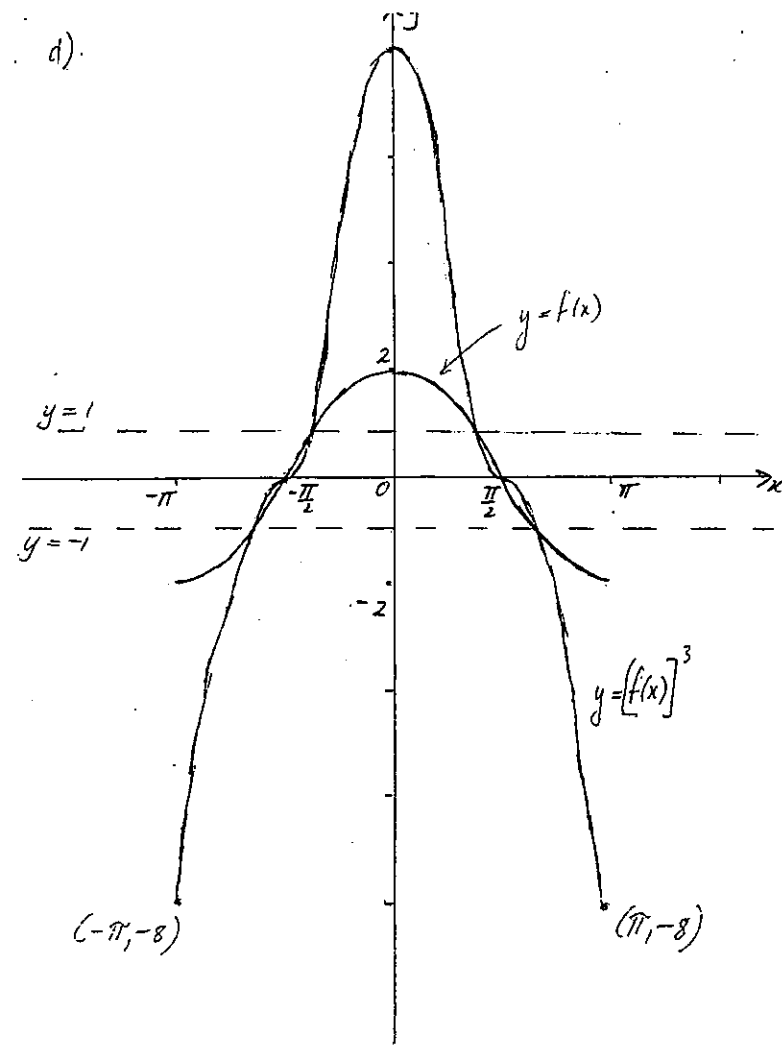
c) (i)



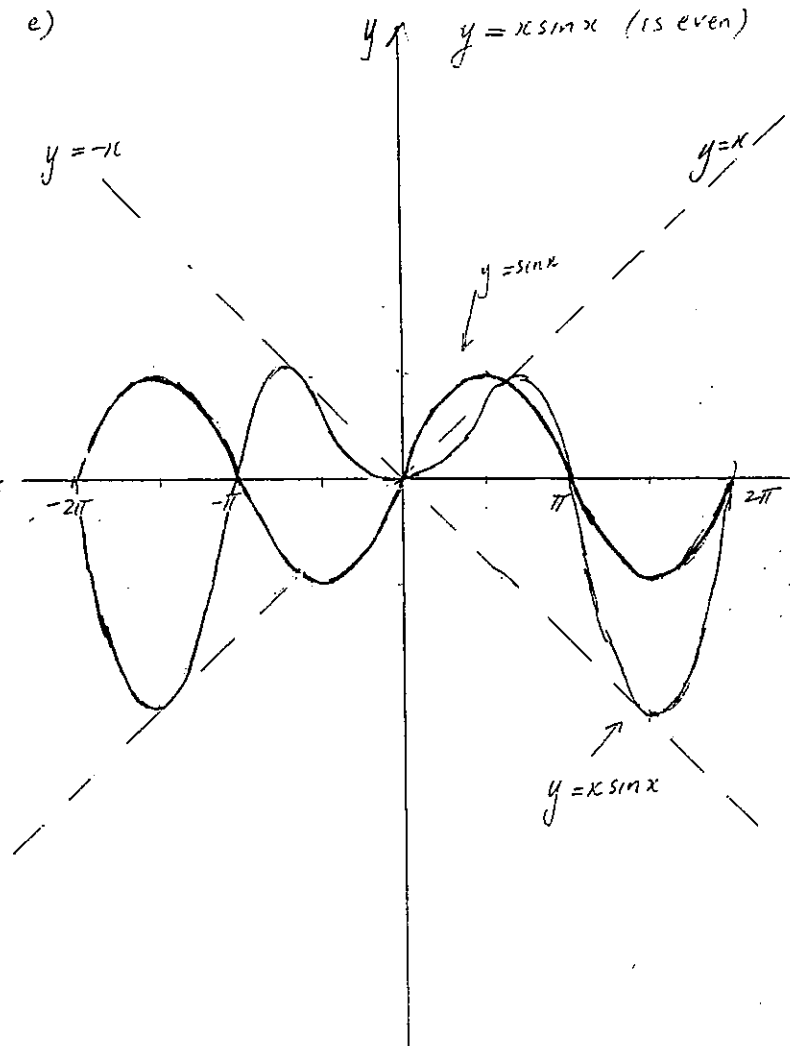
c) (ii)



d)

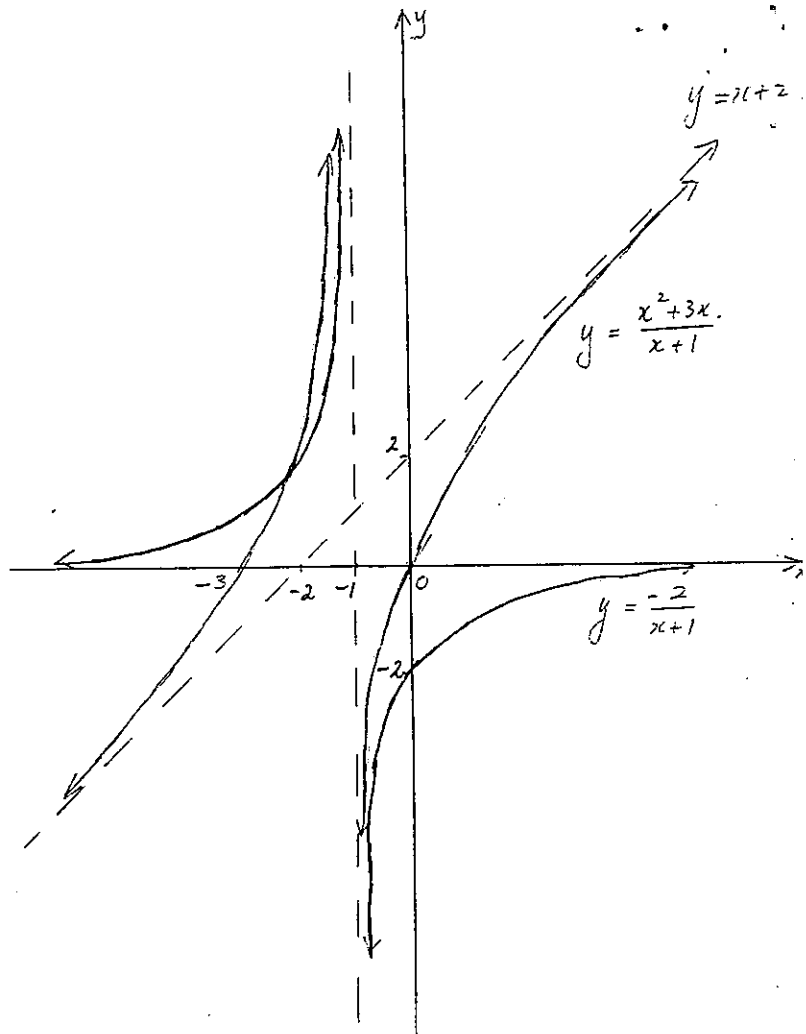


e)



$$f) \frac{x^2 + 3x}{x+1} = x+2 + \frac{-2}{x+1}$$

$$\begin{array}{r} x+2 \\ x+1 \overline{) x^2 + 3x} \\ \underline{x^2 + x} \\ 2x \\ \underline{2x + 2} \\ -2 \end{array}$$



INTEGRATION

$$a) (i) \int \frac{e^{\tan x}}{\cos^2 x} dx = \int \sec^2 x \cdot e^{\tan x} dx = e^{\tan x} + c$$

$$\begin{aligned} (ii) \int \tan^4 x dx &= \int \tan^2 x \cdot \tan^2 x dx \\ &= \int (\sec^2 x - 1) \cdot \tan^2 x dx \\ &= \int (\sec^2 x \tan^2 x - \tan^2 x) dx \\ &= \int [\sec^2 x \tan^2 x - (\sec^2 x - 1)] dx \\ &= \int (\sec^2 x \tan^2 x - \sec^2 x + 1) dx \\ &= \frac{\tan^3 x}{3} - \tan x + x + c \end{aligned}$$

$$(iii) \text{ let } u = x \quad \text{and} \quad \frac{du}{dx} = 1 \quad \text{and} \quad \frac{dv}{dx} = \cos 2x \quad V = \frac{1}{2} \sin 2x$$

$$\begin{aligned} \int x \cdot \cos(2x) dx &= \frac{x}{2} \sin 2x - \frac{1}{2} \int \sin 2x dx \\ &= \frac{x}{2} \sin 2x - \frac{1}{2} \left(-\frac{1}{2} \cos 2x \right) + c \\ &= \frac{x}{2} \sin 2x + \frac{1}{4} \cos 2x + c \end{aligned}$$

$$(iv) \int \frac{dx}{\sqrt{x^2 - 8x + 25}} = \int \frac{dx}{\sqrt{x^2 - 8x + 16 + 9}} = \int \frac{dx}{\sqrt{(x-4)^2 + 9}}$$

$$\text{let } u = x - 4$$

$$\frac{du}{dx} = 1 \rightarrow \frac{dx}{du} = 1$$

$$\therefore \int \frac{dx}{\sqrt{x^2 - 8x + 25}} = \int \frac{1}{\sqrt{u^2 + 9}} \cdot 1 du$$

(iv) continued

$$\begin{aligned}\int \frac{dx}{\sqrt{x^2 - 8x + 25}} &= \log_e \left[u + \sqrt{u^2 + 9} \right] + C \\ &= \log_e \left[(x-4) + \sqrt{(x-4)^2 + 9} \right] + C \\ &= \log_e \left[(x-4) + \sqrt{x^2 - 8x + 25} \right] + C\end{aligned}$$

b) (i) $x = 2 \sin \theta$ when $x = 1$, $\theta = \frac{\pi}{6}$
 $\frac{dx}{d\theta} = 2 \cos \theta$ $x = 0$, $\theta = 0$

$$\begin{aligned}\therefore \int_0^1 \sqrt{4-x^2} dx &= \int_0^{\frac{\pi}{6}} \sqrt{4-4\sin^2 \theta} \cdot 2 \cos \theta d\theta \\ &= \int_0^{\frac{\pi}{6}} 2 \cos \theta \cdot 2 \cos \theta d\theta \\ &= 4 \int_0^{\frac{\pi}{6}} \cos^2 \theta d\theta\end{aligned}$$

b) (ii) (continued)

$$\begin{aligned}\int_2^{10} \frac{x^2 dx}{\sqrt{x-1}} &= 2 \left[3 + 18 + \frac{243}{5} - 1 - \frac{2}{3} - \frac{1}{5} \right] \\ &= 2 \left[\frac{242}{5} + \frac{58}{3} \right] \\ &= \frac{2032}{15}\end{aligned}$$

c) (i)

$$\begin{aligned}\text{R.H.S.} &= \frac{1}{2} \left[\sin A \cos B + \cos A \sin B + \sin A \cos B - \cos A \sin B \right] \\ &= \frac{1}{2} \left[2 \sin A \cos B \right] \\ &= \sin A \cos B \\ &= \text{L.H.S.}\end{aligned}$$

$$\begin{aligned}\text{(ii)} \int_0^{\frac{\pi}{2}} \sin 5x \cos 3x dx &= \frac{1}{2} \int_0^{\frac{\pi}{2}} (\sin 8x + \sin 2x) dx \\ &= \frac{1}{2} \left[-\frac{1}{8} \cos 8x - \frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{2}} \\ &= -\frac{1}{16} \left[\cos 8x + 4 \cos 2x \right]_0^{\frac{\pi}{2}}\end{aligned}$$

$$\text{Now } \cos^2 \theta = \frac{1}{2} (1 + \cos 2\theta)$$

$$\begin{aligned}\therefore \int_0^1 \sqrt{4-x^2} dx &= 2 \int_0^{\frac{\pi}{6}} (1 + \cos 2\theta) d\theta \\ &= 2 \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{6}} \\ &= 2 \left[\frac{\pi}{6} + \frac{1}{2} \times \frac{\sqrt{3}}{2} - 0 \right] \\ &= 2 \left[\frac{\pi}{6} + \frac{\sqrt{3}}{4} \right]\end{aligned}$$

(ii) let $u^2 = x-1$ when $x=10$, $u=3$
 $x = u^2 + 1$ $x=2$, $u=1$
 $\frac{dx}{du} = 2u$

$$\begin{aligned}\therefore \int_2^{10} \frac{x^2 dx}{\sqrt{x-1}} &= \int_1^3 \frac{(u^2+1)^2}{u} \cdot 2u du \\ &= 2 \int_1^3 (u^4 + 2u^2 + 1) du \\ &= 2 \left[\frac{u^5}{5} + \frac{2u^3}{3} + u \right]_1^3\end{aligned}$$

$$\begin{aligned}\therefore \int_0^{\frac{\pi}{2}} \sin 5x \cos 3x dx &= -\frac{1}{16} \left[1 - 4 - (1 + 4) \right] \\ &= -\frac{1}{16} [-8] \\ &= \frac{1}{2}\end{aligned}$$

POLYNOMIALS

$$\begin{aligned} a) (i) \quad x^4 - x^2 - 20 &= (x^2 - 5)(x^2 + 4) \\ &= (x - \sqrt{5})(x + \sqrt{5})(x^2 + 4) \end{aligned}$$

$$(ii) \quad x^4 - x^2 - 20 = (x - \sqrt{5})(x + \sqrt{5})(x + 2i)(x - 2i)$$

$$\begin{aligned} b) \quad S(2+i) &= (2+i)^3 - 2(2+i)^2 - 3(2+i) + 10 \\ &= (2+i)(4-1+4i) - 2(3+4i) - 6-3i + 10 \\ &= (2+i)(3+4i) - 6 - 8i - 6 - 3i + 10 \\ &= 6 + 8i + 3i - 4 - 6 - 8i - 6 - 3i + 10 \\ &= 0 \end{aligned}$$

$\therefore (2+i)$ is a zero.

$\therefore 2-i$ is a zero as zeros of a Real Polynomial appear in conjugate pairs.

$$\therefore P(1) = 0 = P'(1) = P''(1).$$

$\therefore x=1$ is (only) root of multiplicity 3

i.e. $(x-1)^3$ is a factor.

$$\therefore P(x) = (x-1)^3(x-3) \text{ By Inspection.}$$

$\therefore$ Roots of $P(x)$ are 1, 3

d) α is a root.

$$\therefore \alpha^3 - 5\alpha^2 + 8\alpha + 24 = 0$$

We want $x = \alpha^2$ to be a root.

$$\therefore \alpha = \pm\sqrt{x}$$

$$\therefore (\pm\sqrt{x})^3 - 5(\pm\sqrt{x})^2 + 8(\pm\sqrt{x}) + 24 = 0$$

$$(\pm\sqrt{x})^3 + 8(\pm\sqrt{x}) = 5x - 24$$

$$\pm\sqrt{x} [x + 8] = 5x - 24.$$

Squaring both sides

$$x(x+8)^2 = (5x-24)^2$$

$$x(x^2 + 16x + 64) = 25x^2 - 240x + 576$$

$$\text{Sum of Roots} = -\frac{b}{a} \dots$$

$$(2+i) + (2-i) + \alpha = 2$$

$$\alpha = -2$$

$\therefore$ Roots are $(2+i)$, $(2-i)$ & -2

$$c) \text{ Let } P(x) = x^4 - 6x^3 + 12x^2 - 10x + 3$$

$$P'(x) = 4x^3 - 18x^2 + 24x - 10$$

$$P''(x) = 12x^2 - 36x + 24$$

For a root of multiplicity of 3

$$P''(x) = P'(x) = P(x) = 0$$

$$\therefore 12x^2 - 36x + 24 = 0$$

$$x^2 - 3x + 2 = 0$$

$$(x-2)(x-1) = 0$$

$$x = 2, 1$$

$$P'(1) = 4 - 18 + 24 - 10$$

$$= 0$$

$$P(1) = 1 - 6 + 12 - 10 + 3$$

d) (continued)

$$x^3 + 16x^2 + 64x = 25x^2 - 240x + 576$$

$$\therefore x^3 - 9x^2 - 304x - 576 = 0$$

$$e) \text{ Let } \frac{2x^2 + x + 5}{(x-3)(x^2+4)} = \frac{a}{x-3} + \frac{bx+c}{x^2+4}$$

$$\therefore 2x^2 + x + 5 = ax^2 + 4a + (x-3)(bx+c)$$

$$= ax^2 + 4a + bx^2 + cx - 3bx - 3c$$

$$= (a+b)x^2 + (c-3b)x + 4a-3c$$

$\therefore$ Equating Coefficients

$$a+b = 2 \quad ; \quad c-3b = 1 \quad ; \quad 4a-3c =$$

$$3a+3b = 6 \quad +$$

$$c-3b = 1$$

$$3a+c = 7 \longrightarrow 9a+3c = 21$$

$$4a-3c = 5$$

$$13a = 26$$

$$a = 2$$

$$\therefore b = 0 \text{ and } c = 1$$

e) (continued)

$$\frac{2x^2 + x + 5}{(x-3)(x^2+4)} = \frac{2}{x-3} + \frac{1}{x^2+4}$$

f) Possible rational zeros are
 $\pm 1, \pm \frac{1}{2}, \pm 3, \pm \frac{3}{2}$

$$\phi\left(\frac{1}{2}\right) \neq 0, \phi\left(-\frac{1}{2}\right) \neq 0$$

$$\phi\left(\frac{3}{2}\right) = \frac{27}{4} - \frac{45}{4} + \frac{6}{4} + \frac{12}{4} = 0$$

$\therefore x = \frac{3}{2}$ is a root

$(2x-3)$ is a factor of $\phi(x)$

$$\therefore \phi(x) = (2x-3)(x^2+ax-1)$$

$$= 2x^3 + 2ax^2 - 2x - 3x^2 - 3ax + 3$$

$$= 2x^3 + (2a-3)x^2 - (2+3a)x + 3$$

$$\therefore 2a-3 = -5 \quad \text{Equating coefficients}$$

$$a = -1$$

$$\therefore \phi(x) = (2x-3)(x^2-x-1)$$

$$\therefore \text{Zeros are } \frac{3}{2}, \frac{1 \pm \sqrt{5}}{2}$$

$$g)(i) \quad z = \cos w + i \sin w$$

$$z^{-1} = (\cos w + i \sin w)^{-1}$$

$$= \cos(-w) + i \sin(-w)$$

$$z^{-1} = \cos w - i \sin w$$

$$\therefore z + z^{-1} = 2 \cos w$$

$$z^n = (\cos w + i \sin w)^n$$

$$= \cos(nw) + i \sin(nw)$$

$$z^{-n} = \cos(-nw) + i \sin(-nw)$$

$$= \cos(nw) - i \sin(nw)$$

$$\therefore z^n + z^{-n} = 2 \cos(nw)$$

$$(ii) \quad \left(z + \frac{1}{z}\right)^4 = z^4 + 4z^2 + 6 + \frac{4}{z^2} + \frac{1}{z^4}$$

$$\therefore (2 \cos w)^4 = z^4 + \frac{1}{z^4} + 4\left(z^2 + \frac{1}{z^2}\right) + 6$$

$$= 2 \cos(4w) + 4 \times 2 \cos(2w) + 6$$

$$16 \cos^4 w = 2 \cos(4w) + 8 \cos(2w) + 6$$

$$\cos^4 w = \frac{1}{8} \cos(4w) + \frac{1}{2} \cos(2w) + \frac{3}{8}$$